

BI PRACTICAL ASSESSMENT

Power BI, SQL & Telecom Business Analytics

Take-home practical assessment

1. Assessment Overview

You are provided with a sample telecom dataset containing customer, subscription, consumption, calls, payments and mobile-app activity information. Your task is to build an analytical solution that can be used by management to understand customer behavior, value, engagement and business opportunities.

2. Data Available

Data Area	Typical Analytical Use
Customer	Customer base, segmentation, customer profiling
Subscription / Plan	Plan mix, customer value, plan comparison
Consumption	Usage, trends, usage per customer
Calls	Engagement, behavior comparison
Payments	Customer value, payment behavior, monetization
Mobile App	Digital engagement and behavioral analysis

To download the data on csv format, please use this URL [CSV Files](#)

3. General Requirements

- Use the supplied dataset as the source of truth. Do not manually change source records.
- All transformations and calculations should be reproducible after refresh.
- Build the Power BI model so that the same customer is not counted multiple times because of fact-to-fact joins.
- Use measures for analytical KPIs where appropriate and keep business definitions clear.
- Your report should tell a business story. Do not build a collection of unrelated charts.
- If the available data is not sufficient to prove a business conclusion, clearly state the limitation and what additional data would be required.
- Submit the PBIX file and the SQL script used for the SQL tasks. Include a short README with assumptions and the main insights.

4. Business Context

Assume you are supporting a telecom management team. Management is interested not only in how many customers and how much usage exists, but also in customer value, engagement, plan behavior, opportunities for growth, and potential risks. Your analysis should help management decide where to focus.

PRACTICAL TASKS

Complete all tasks using the supplied dataset.

A. SQL

1. Customer-Level Multi-Fact Analysis

Create a SQL result with exactly one row per customer containing: Total Consumption, Total Payments and Total Calls. Aggregate each activity to customer level before combining the results. The final output must not contain duplicated customers or inflated totals.

2. SQL Optimization

Select a query from Task 1 (or a similarly complex query) that has a genuine performance problem on the full data volume — not a trivial query. Provide an optimized version and identify at least three concrete changes you made (e.g., join order, indexing/partitioning, avoiding repeated table scans). Show or describe the execution plan difference before and after, and explain how you verified the optimized query returns the same result.

B. POWER BI MODEL & DAX

3. Analytical Data Model

Build a Power BI model suitable for ongoing reporting. Clearly separate descriptive customer/plan/date information from transactional or activity data. Define the grain of the main activity tables, create appropriate relationships, and ensure that filtering a customer, plan or date produces correct results without double counting.

4. Core KPI & DAX Layer

Create the reusable KPI/DAX layer you believe an ongoing telecom management report needs. At minimum, include measures for Total Customers and Active Customers (other tasks depend on these — document the business definition you use for Active Customer). Beyond that, decide which additional measures matter (e.g., value, consumption, trend, or share-based measures) and briefly justify your choices in the README. Any Percentage-of-Total style measure you build must work correctly when users filter by Subscription Plan.

C. BUSINESS INSIGHTS & EXECUTIVE POWER BI

5. Executive Dashboard

Build an Executive Overview that allows management to understand the customer situation quickly: customer base, customer value, engagement, trends/change, and major contributing plans or segments. Add supporting analytical pages that let the user investigate the drivers behind the headline results.

6. Business Insights & Opportunities

From your dashboard, pull out THREE business insights that matter to telecom management — at least one growth opportunity and at least one risk. For each: state the finding, back it with at least two numbers/KPIs, give the most likely driver, and recommend one concrete action. 'Plan A has the most customers' is not an insight — say what it means and what to do about it.

7. Customer Segmentation

Create at least FOUR customer segments based on measurable behavior from the dataset, such as consumption, calls, payments, mobile-app activity or subscription plan. For each segment, show customer count, share of the customer base, value/consumption per customer, an engagement indicator, a clear business interpretation and a recommended action. Explain why your segmentation is useful for telecom decision-making.

D. AI / ML

8. AI / ML Use Case

Propose ONE AI/ML use case that could realistically be built using this dataset. Define the prediction target, 5–8 candidate features, how you would split training and testing data, two appropriate evaluation metrics, one way to prevent data leakage, how the prediction would be presented or used in Power BI, and how you would monitor model/data drift. Explain why a model prediction should support — not automatically replace — a business decision.

Final Deliverable

Your final submission should contain: (1) PBIX file, (2) SQL script, and (3) a short README summarizing assumptions, the KPI choices you made in Task 4, the main business insights, important limitations, and recommended actions. In the README, also include a few sentences on how you would explain an inconclusive finding (e.g., the driver behind a consumption drop) to a non-technical manager.